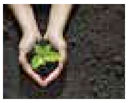

GSMA standard security
Thales, Telstra, Microsoft and Arduino are working together to implement a secure GSMA solution. **https://dgit.in/apr20-77**

How safe are IoT devices?
A report has found that a staggering 57 percent of IoT devices are susceptible to severe attacks. **https://dgit.in/apr20-78**

GOING HARD IN SFF?

Agent001's hard tubing is attracting too much attention.

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While hanging out with my buddies the other day, one of the guys was overly engrossed with his computer screen and all we could hear from him were a few gasps and groans every now and then. Naturally, we were extremely curious as to what the dude was up to and happened to casually sneak a glance at his screen while walking by. Turns out he was checking out some outrageously cool hard tube custom liquid cooled PCs. I can understand the effect that a gorgeously made computer can have on any person. People have their own preferences. Some like a classy design with straight lines and a dash of flourish whereas some folks prefer no flash and just like to get down to business. And then are the folks who like to see all sorts of twists and curves along with some flattering RGB lighting. We don't look down upon anyone aspiring to get themselves one of these beauties but some of these are truly high maintenance configurations that can be just as tedious to maintain as they are enchanting to look at.

After a while he started wondering whether he should go for a hard tube setup and wanted to know my opinion knowing well that I was nearly done building a custom loop for my own machine. Getting parts for a hard tube custom liquid cooling loop was quite the chore until last year since we had little options in the market. We could find kits from EK and Thermaltake which meant that you were limited to building what could be possible using the small set of hardware available. EK kits continue to be very limited in options whereas Thermaltake has opened up most of their portfolio in India. Corsair is the latest to bring their Hydro-X series to India, although their offerings are a little limited they do provide the option to buy each component individually and don't offer kits that restrict your creativity. So for someone looking to build a hard tube or soft tube custom loop, you now have access to most of the top brands. Plus independent distributors around the country have started importing and offering components from brands such as Bitspower, Bykski, Barrow and other OEMs. You can also get the same via AliExpress if you know how to get the shipment sorted or else you'll only get your orders after 30-60 days.

Coming back to my friend's query, we started going through his current configuration. For starters, he has a small form factor chassis with a high-end CPU and GPU. Not that it is impossible to install a custom loop in a small form factor chassis, getting the right sized components becomes essential since space is a serious constraint. Then there's the fact that the reduced space makes it difficult to fit the right sized radiators. In most of these small form factor cases you can only squeeze in a 240mm radiator which is sufficient for cooling a CPU or even a GPU in combination but you'll be stressing the system. If there isn't enough air-flow then the loop temperature will be too high and you would have spent a ridiculous amount for no benefit. Moreover, hot loops are bad for the tubes as well as for the gasket rings used in the fittings. PETG loops are better when the coolant temperature is 40 degree celsius or below, and for warmer climates, it is better to switch to PMMA tubing which can handle up to 60 celsius without issues. We ruled out using a custom hard tube loop system in his small form factor machine since it didn't have enough space to fit the right size of radiators. Especially in India where summer temps can only aggravate things.

In the end it was decided that my friend should either get a bigger case if he really wanted to install a custom loop or he could stick with his current setup and perhaps explore an AIO for his CPU if he really needed to reduce the temperature inside his chassis.